



Planet SkySat Collect: Quality Processes Review

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1 Introduction

1.1 Scope

This document summarises a Data Product Quality Processes Review for the Planet SkySat Collect data product as part of the EODH Data Product Quality Process [AD-1]. The purpose is to demonstrate that the product has been developed and generated following processes that adhere to best practices.

To achieve this, the EODH Data Product QA Process adheres to the ESA/NASA EO Product QA Framework [RD-1]. This framework defines requirements for a variety of aspects of Data Product quality – awarding grades of *Basic*, *Good*, *Excellent*, or *Ideal*. Such assessments are reported with a maturity matrix – this is presented in Section 2. The rest of the document provides the underpinning evidence justifying the awarded grades. The assessment has been completed by the EODH QA Service on behalf of the Data Product provider.

The result of this evaluation, referred to as a *QA Check Result*, will be added as an annotation to the EODH Catalogue for user interaction.

1.2 Changes Log

Issue	Date	Description
0.1	17/12/2025	Version for internal review.
1.0	22/02/2026	Draft for pre-publication review.

1.3 Applicable Documents

[AD-1] EODH QA Service, ‘Data Product Quality Assurance Process’, 2025.

1.4 Reference Documents

[RD-1] Hunt et al., ‘Earth Observation Mission Quality Assessment Framework’ v2.2, EDAP Project Document, 2021. [Available Online](#).

[RD-2] Planet L1 Data Quality Q2 2025 [Report](#), Status of Calibration and Data Quality for the SkySat Constellation.

[RD-3] Understanding SkySat Instruments: [SkySat | Planet Documentation](#)

[RD-4] Planet Imagery [Product Specifications](#), December 2023.

[RD-5] On-Orbit [Radiometric Calibration](#) of the Planet Satellite Fleet.

[RD-6] [OGC\(r\) GML 3.1.1 Application Schema](#) for Earth Observation Products

[RD-7] JCGM, “Evaluation of measurement data - Guide to the expression of uncertainty in measurement,” *JCGM*, vol. 100, 2008, Accessed: Feb. 06, 2018. [\[Online\]](#)

1.5 Acronyms & Abbreviations

EDAP	Earthnet Data Assessment Project
EO	Earth Observation
EODH	Earth Observation Data Hub
ESA	European Space Agency
NASA	National Aeronautics and Space Administration
NCEO	National Centre for Earth Observation
NPL	National Physical Laboratory
QA	Quality Assurance
DN	Digital Number
DEM	Digital Elevation Model
GCP	Ground Control Point
PICS	Pseudo Invariant Calibration Point

2 Review Summary – Product Maturity Matrix

The *Data Product Quality Processes Review* is aimed at demonstrating that a given Data Product is developed and generated following processes that adhere to best practices.

To achieve this, the EODH *Data Product QA Process* adheres to the ESA/NASA EO Product QA Framework [RD-1]. Such assessments are reported in a maturity matrix style, where each box represents an area of assessment for which underpinning requirements are defined – the box colour represents the achieved grade, *Basic*, *Good*, *Excellent*, or *Ideal*. Additionally:

- *Not Assessed* means this aspect of data quality was not assessed at this time.
- Not Assessable means the assessment of this aspect of data quality was not possible due to a lack of available evidence.
- The padlock symbol means that the supporting evidence reviewed is not publicly available.

The *Data Product Quality Processes Review* for Planet SkySat is presented in Figure 1. The rest of the document provides the underpinning evidence justifying the awarded grades, as follows:

- Section 3 provides the evidence for the Product Information column.
- Section 4 provides the evidence for the Metrology column.
- Section 5 provides the evidence for the Product Generation column.

Data Product Quality Processes Review			Key
Product Information	Metrology	Product Generation	
Product Details	Sensor Calibration & Characterisation	Calibration Algorithm	Not Assessed
Availability & Accessibility	Geometric Calibration & Characterisation	Geometric Processing	Not Assessable
Product Format, Flags & Metadata	Metrological Traceability Documentation	Product Specific Processing	Basic
User Documentation	Uncertainty Characterisation		Good
	Ancillary Data		Excellent
			Ideal
			Not Public

Figure 1 - Data Product Quality Processes Review Maturity Matrix for Planet SkySat

3 Product Information

Product Details	
Grade: Excellent	
Justification	<i>Almost all required information available publicly, however not all in the same location.</i> <i>Information found in references: [RD-2], [RD-3], [RD-4], [RD-5]</i>
Product Name	<i>SkySatCollect</i>
Sensor Name	<i>SkySat Camera</i>
Sensor Type	<i>CMOS Frame Camera with Panchromatic and Multispectral (Blue, Green, Red, NIR) halves.</i>
Mission Type	<i>Constellation – 21 satellites</i>
Mission Orbit	<i>Morning & Afternoon Sun Synchronous</i>
Product Version Number	<i>Processing versions included in the product metadata.</i>
Product ID	<i>{acquisition_date}_{acquisition_time}_{satellite_id}_{frame_id}_{bandProduct}.{extension}</i> <i>Example: 20200815_091045_ssc6_u0002_visual.tif</i>
Processing level of product	<i>SkySat Collect is a collection of imagery products, or “items”. Each “item” has a list of “assets” that can be derived from the source data available for download. Below are the lists of “assets” available for the SkySat collect product:</i> <i>Basic data</i> <i>basic_l1a_all_frames: all basic l1a panchromatic frames and accompanying RPCs and pinhole json files that make up the collect.</i> <i>Orthorectified Multispectral Products</i> <i>ortho_analytic: orthorectified 16-bit 4-Band radiometrically corrected image.</i> <i>ortho_analytic_dn: orthorectified 16-bit 4-Band DN Image.</i> <i>ortho_analytic_sr: surface reflectance product-orthorectified, radiometrically calibrated and atmospherically corrected.</i> <i>Masks</i> <i>ortho_analytic_udm: orthorectified 16-bit 4-Band DN Image Unusable Data Mask.</i> <i>ortho_analytic_udm2: orthorectified usable data mask (Cloud 2.0)</i> <i>Panchromatic Products</i>

	• ortho_panchromatic: orthorectified Radiometrically-calibrated panchromatic image stored as 16-bit scaled radiance. • ortho_panchromatic_dn: orthorectified 16-bit 1-band Image Pansharpened Products • ortho_pansharpended: orthorectified 16-bit 4-band Pansharpened Image. Visual Product: • ortho_visual: visual image with colour-correction.
Measured Quantity Name	1. TOA Radiance: <i>ortho_analytic, ortho_panchromatic, ortho_pansharpended</i> 2. DN: <i>basic_l1a_all_frames, ortho_analytic_dn, ortho_panchromatic_dn</i> 3. BOA Reflectance: <i>ortho_analytic_sr</i>
Measured Quantity Units	1. $1/100 \text{ W sr}^{-1} \text{ m}^{-2} \mu\text{m}^{-1}$ 2. [-] 3. [-]
Stated Measurement Quality	Ortho products stated to have $\pm 5\%$ relative radiometric accuracy at < 10 degrees off-nadir angle for the analytic, panchromatic product. From latest (Q2 2025) Quality report, the absolute geolocation in the 90th percentile (m): • Ortho products: 6.65 m • Basic products: 199.975 m * With the relative geolocational accuracy (scene-to-scene) from the quarterly report given as 0.58 m in the 90th percentile. * Note: Basic products are uncalibrated and in a raw digital number format with no geometric calibration applied. These products are for use by users with specific understanding of how to process the data for their needs; average users should use Ortho products.
Spatial Resolution	0.597m at the equator, with pansharpended image at 0.5m.
Spatial Coverage	$\pm 80^\circ$ (dependent on season) Approximate 5.73 km swath.
Temporal Resolution	4-5 days (per satellite) at altitude of 500km
Temporal Coverage	First image in July 2014.
Point of Contact	Planet
Product locator (DOI/URL)	Available for purchase through the Planet website
Conditions for access and use	Requires commercial licensing
Limitations on public access	Yes (see above)
	Extracts from Planet content online:

Product Abstract	<i>A SkySat Collect Product is created by composing many SkySat Scenes along an imaging strip into an orthorectified segment. Collects are often composed of 60 or more scenes and will be at least 5 km long. They are represented in the Planet Platform as the SkySatCollect item type. This product may be easier to handle, if you're looking at larger areas of interest with SkySat imagery. Due to the image rectification process involved in creating this product, Collect is generally recommended over the Scene product when the AOI spans multiple scenes, particularly if a mosaic or composite image of the individual scenes is required. Collect performs necessary rectification steps automatically. This is especially useful for users who don't feel comfortable doing orthorectification manually.</i> <i>SkySat, operated by Planet, is a high-resolution constellation of 21 satellites, able to image revisit any location on Earth up to 10x daily (a daily collection capacity of 4,000 km²/day). SkySat supports the Planet high resolution tasking, which allows you to choose when and where to capture SkySat imagery. SkySat produces 4-band and panchromatic imagery and is sampled at 50 centimetres per pixel when orthorectified.</i> <i>The SkySat satellite constellation consists of multiple launches of the Planet SkySat-C generation satellites, first launched in 2016. Each satellite is 3-axis stabilized and agile enough to slew between different targets of interest. Each satellite has four thrusters for orbital control, along with four reaction wheels and three magnetic torquers for attitude control. Currently, the SkySat constellation consists of morning and afternoon sun-synchronous satellites with an operational altitude of 475 km. The minimum and maximum latitude coverage is up to $\pm 80^\circ$ but depends on season. SkySat collects approximately a 5.73 km swath.</i> <i>All SkySat satellites contain Cassegrain telescopes with a focal length of 3.6 m, with three overlapping 5.5-megapixel CMOS imaging detectors making up the focal plane. The central detector is slightly offset from the side detectors, giving in SkySat imagery it's unique "tuning fork" outline. SkySat can collect frame imagery, stereo imagery, and video in day or night collects. SkySat images are typically collected in short exposures, coregistered, and stacked to improve the SNR and GSD with super-resolution. SkySat collects RGB, NIR and panchromatic imagery.</i>
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Availability & Accessibility	
Grade: Basic	
Justification	<i>Good accessibility and availability, meeting many of the FAIR principles, however it does not comply with FAIR.</i> <i>Description of data management plan is not publicly available.</i>
Compliant with FAIR principles	No.
Data Management Plan	Not known.
Availability Status	Commercially available.

Product Format	
Grade: Good	
Justification	<i>Metadata provides appropriate breadth of information for each data product.</i> <i>Does not follow CEOS-ARD, but convention is documented.</i>
Product File Format	<i>L1C: Data stored as .tiff file, and metadata provided in a .xml file. Informative file naming, although not clear if this is standard.</i>
Metadata Conventions	<i>XML metadata files follow [RD-6] specifications.</i>
Analysis Ready Data?	No

User Documentation		
Grade: Good		
Justification	<i>Several technical documents available for users, including information on the radiometric calibration.</i> <i>Up-to-date information can usually be found via the Planet website, although found in various places.</i>	
Document	Reference	QA4ECV Compliant
Product User Guide	[RD-3]	No
ATBD	No formal ATBD	No
Radiometric Calibration Information	[RD-2]	No

4 Metrology

Sensor Calibration & Characterisation	
Grade: Good	
Justification	<i>Information into the in-orbit calibration is documented.</i> <i>Quarterly results are published following regular characterisation methods, with calibration coefficients regularly monitored and updated with on-orbit calibration techniques.</i> <i>No information or relevant documentation was found regarding the pre-flight calibration.</i>
References	<i>[RD-2], [RD-5]</i>

Geometric Calibration & Characterisation	
Grade: Good	
Justification	<i>Information on post-flight geometric calibration available, using DEMs & GCPs, with geometric performance reported quarterly.</i> <i>Ground Control Points (GCPs) are used in the creation of every image.</i>
References	<i>[RD-2]</i>

Metrological Traceability Documentation	
Grade: Not assessable	
Justification	<i>Metrology Traceability Chain has not been documented.</i>
References	

Uncertainty Characterisation	
Grade: Basic	
Justification	<i>A proxy for the radiometric uncertainty is calculated and published in a quarterly report.</i> <i>This proxy is calculated through vicarious calibration at the RadCalNet sites to achieve an absolute radiometric link between corrected satellite DN and physical radiance unit.</i> <i>Uncertainty in calibration given from 272 scenes in quarterly report.</i>

	<i>Uncertainty values not given per data product.</i> <i>Accuracy of geometric performance comprehensively evaluated for a subset of data against a reference dataset and reported.</i> <i>Uncertainty evaluation does not follow GUM approach.</i>
References	<i>[RD-2], [RD-4], [RD-5], [RD-7]</i>

Ancillary Data	
Grade: Good	
Justification	<i>The primary ancillary datasets (DEM) used for orthorectification in each region stated in quarterly report.</i> <i>Atmospheric correction done using MODIS/VIIRS input data where possible. Where MODIS/VIIRS data is unavailable, a default AOD value is used, along with a predefined 6SV internal model.</i>
References	<i>[RD-2], [RD-3], [RD-4]</i>

5 Product Generation

Calibration Algorithm	
Grade: Excellent	
Justification	<i>A full and detailed description of the on-orbit calibration algorithm processing carried out, outlining method from pixel DN values to Radiance, with ATBD-like information.</i> <i>Algorithm appears fit-for-purpose in terms of the mission's stated performance for all expected use cases.</i>
References	<i>[RD-2], [RD-4], [RD-5]</i>

Geometric Processing	
Grade: Good	
Justification	<i>As above, processing steps provided, along with reference datasets used. There is sufficient confidence in the calibration quality with the geometric accuracy provided.</i> <i>Certain processes such as the orthorectification appear fit-for-purpose but lack detailed documentation, other geometric processing steps have little information.</i>
References	<i>[RD-2], [RD-4], [RD-5]</i>

Product-Specific Processing	
Grade: Good	
Orthorectification	
Justification	<i>Geometric accuracy is stated for ortho products in the quarterly report and the process is well-documented and considered fit for purpose.</i> <i>Steps to removes terrain distortions are clearly defined as follows:</i> <i>1. Tie points are matched between source images and reference datasets (e.g., ALOS, NAIP, Landsat) to generate RPCs.</i> <i>2. Scenes are then geometrically corrected using RPCs and terrain models (from sources like Intermap, NED, SRTM), with elevation data snapshots archived for traceability.</i> <i>The above processing applies for the following "assets":</i> <i>ortho_analytic, ortho_analytic_dn, ortho_analytic_sr, ortho_panchromatic, ortho_panchromatic_dn, ortho_pansharpened, ortho_visual</i>

Reference	[RD-3], [RD-4]
Visual Product Processing	
Justification	<i>Visual products are described to present the imagery as natural color, optimizing colors as seen by the human eye, with custom color curves applied post warping to deliver a visually appealing image.</i> <i>This process is not as well-documented as the corresponding for other Planet products (PlanetScope & RapidEye) in the same report and therefore needs more information in order to be considered fit for purpose.</i> <i>Visual product processing done only on the ortho_visual “asset”.</i>
Reference	[RD-4]
Pansharpening	
Justification	<i>Pansharpening is mentioned in the Product Specification document as a process for these “assets” but does not explicitly describe how it is carried out. Therefore, more information is needed in order to deem it fit for purpose.</i> <i>However, anomalies that arise from the process, such as “oversharpening”, are described in quarterly report.</i> <i>The above processing applies for the following “assets”: ortho_pansharpened, ortho_visual</i>
Reference	[RD-2], [RD-3], [RD-4]